

WTC2005-64050

FORMS OF STATIC FRICTION, SLIDING FRICTION AND ROLLING FRICTION

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ABSTRACT

Although, it seems that the concepts of static friction, sliding friction and rolling friction are common knowledges, their real sources are ironically not so clear. Inappropriate descriptions of them may have hampered the theoretical analyses and calculations of the frictions and deterred their applications in real world. In this paper, we are proposing a new method to model the static friction, sliding friction and rolling friction by three mechanical elements: spring, translatory damper and rotary damper respectively. It is shown that proper descriptions of the frictions, e.g. rolling friction, can help in correcting discrepancies in previous research works.

Key words: static friction, rolling friction, sliding friction

1. INTRODUCTION

There are three kinds of friction forces acting on a ball when it contacts with a plane. They are the sliding, static and rolling frictions. Studies on sliding and static friction had started by da Vinci and Amontons as early as 300 years ago. However, there are still many unknowns about their sources. Rolling friction has been observed for many years, while their mechanism is still not very clear. Many scientists have done some researches on rolling friction in recent years [1-7]. However, inappropriate descriptions of rolling friction still exist, which hamper the theoretical analyses and calculations of rolling friction. In many text books, rolling friction was assumed to take the same form as sliding and static frictions. Some engineers even used tangential forces to describe and calculate the rolling friction, which produced inaccurate results. To prevent errors induced by inappropriate descriptions of frictions, this work is presenting correct forms of the static friction, sliding friction and rolling friction, by which features of the three frictions are distinguished.

2. PROBLEMS OF PREVIOUS FORMS

When taking rolling friction as a tangential force, the frictions between two rotating identical balls are shown in Fig.1. According to Newton's Third Law, $F_{1,2}$ and $F_{2,1}$ are in opposite directions. Hence, the torsional resistance M_{r1} and M_{r2} caused by the rolling friction are in the same direction, which is apparently inaccurate, since rotational resistance due to rolling friction should be in opposite directions of rotating speed.

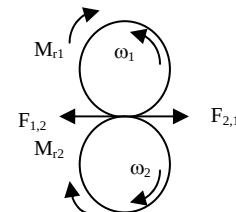


Fig.1 Inaccurate form of rolling friction

3. CORRECT FORMS OF FRICTIONS

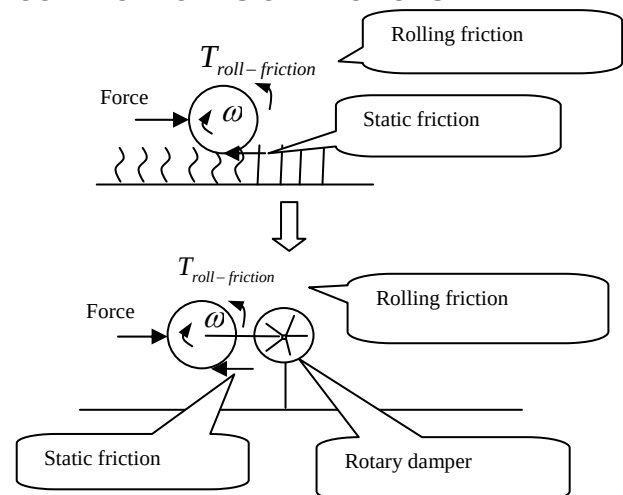


Fig.2 Form of rolling friction

The form of rolling friction is a moment or torsional force acting at the center of the rolling object, reducing the kinetic energy of the moving body. It can be modeled with a rotary damper at the center of the rolling object as shown in Fig. 2. The feature of rolling friction is that kinetic energy is dissipated into heat or other kinds of unrecoverable energy. Adhesion, electrostatic interaction and viscous deformation may be sources of rolling friction.

The form of static friction is a translatory force acting at the point of contact, never reducing or producing kinetic energy. It could be modeled by a spring at the point of contact. The feature of static friction is that, when a force is applied, a potential energy is generated and stored, this energy is recovered later. So, static friction does not change the total

kinetic energy. Elastic deformation may be one of the sources of static friction [8].

The form of sliding friction is a translatory force acting at the point of contact, reducing the kinetic energy of moving body. It could be modeled by a translatory damper at the point of contact. The feature of sliding friction is that kinetic energy is dissipated into heat or other kinds of unrecoverable energy due to sliding motion between surfaces. Adhesion, electrostatic interaction, rupture propagation, viscous deformation of asperities may be sources of sliding friction.

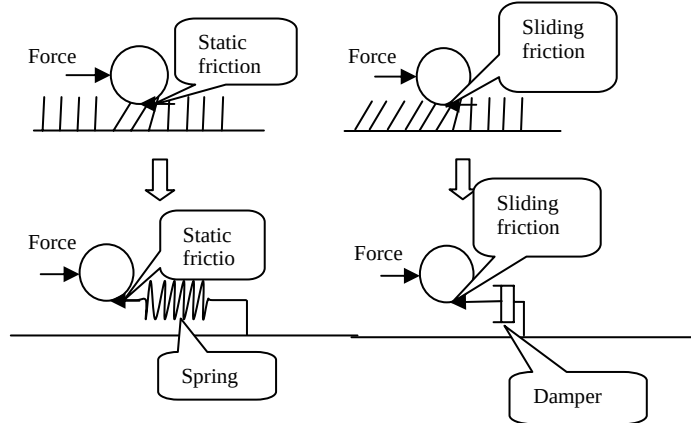


Fig.3 Form of static friction

Fig. 4 Form of sliding friction

3. CASE STUDIES

1) Rolling without slipping

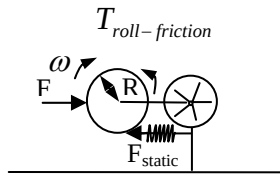


Fig.5 Model of rolling without slipping

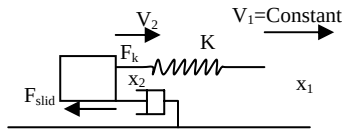


Fig.6 Stick-slip model

When a ball is rolling on a surface without slipping, it is similar to the case of rack-pinion assembly where translatory kinetic energy is converted into rotary kinetic energy. Here, the conversion is made by the static friction.

Although static friction is modeled by spring, the rolling movement makes the contact point changes continuously. So the spring is compressed and released repeatedly. Due to the replacement, no oscillations are observed, since infinite series of oscillations form a continuous motion (similar to the invert of Fourier Series expansion).

Equations of motion are:

$$F_{static} R - T_{roll - friction} = I \ddot{\theta} = I \frac{\ddot{x}}{R}$$

$$F - F_{static} = m \ddot{x}$$

Decrease rate of kinetic energy:

$$\dot{E}_k = \dot{E}_{KT} + \dot{E}_{KR} = (\frac{1}{2} m \dot{x}^2)' + (\frac{1}{2} I \dot{\theta}^2)' = F \dot{x} - T_{roll - friction} \dot{\theta}$$

The equation explains why static friction doesn't affect total kinetic energy. Only rolling friction dissipate kinetic energy into heat.

2) Stick slip phenomenon

Stick-slip is a phenomenon consists of sticking and sliding motions alternatively, where sticking is the occasion when static friction takes effect, sliding is when sliding friction dominates. The phenomena can be described by two alternating models, one moment is static friction model and another is sliding friction model.

Assuming the sliding friction force F_{slid} is constant, with initial conditions: $x_2(0) = 0$ and $\dot{x}_2(0) = 0$, solving equation of motion: $F_k - F_{slid} = m \ddot{x}_2$, we have the displacement and velocity profile of the object as shown in Fig.7.

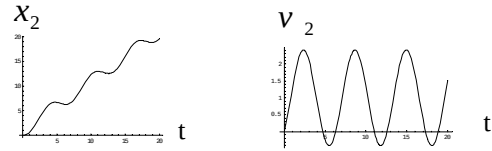


Fig. 7 Displacement and velocity profile in stick-slip

The above figures are different from the phenomena observed in experiment[9], since the models applied are simplified models: (1) the static friction is assumed to be a perfect linear spring, which may not be true; (2) the sliding friction is assumed constant, which may be inaccurate either.

More accurate models could be achieved by obtaining functions $\mu_{slid} = f(v, t)$ and $k = g(v, t)$

In conclusion, the clarification of the forms of static friction, sliding friction and rolling friction can make the calculations of frictions related phenomena more accurate.

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